

Physics 11 Unit Review:

Energy

Name _____

Date _____ Block ____

- Consider task of moving a 250 kg piano from ground level 1.50 meters up into a stationary van. Van has a frictionless ramp that is 8.00 m long.
 - find work needed to lift piano straight up at constant speed
 - find force needed to push piano up ramp at constant speed using Force components vs. work
 - How much work is needed to raise a 3.5 kg object from a shelf that is 1.45 m above ground to a shelf that is 2.50 m above ground?
 - A 200 kg box is pushed with a force of 1100 N for 12 m up a ramp from rest on ground to a height of 6.0 m. Find speed of box at this height.
 - A person pulls on a 20.0 kg lawnmower with a force of 25.0 N exerted on a handle angle of 15.0° above horizontal, moves mower from rest 4.50 m horizontally along a surface with $\mu=0.100$.

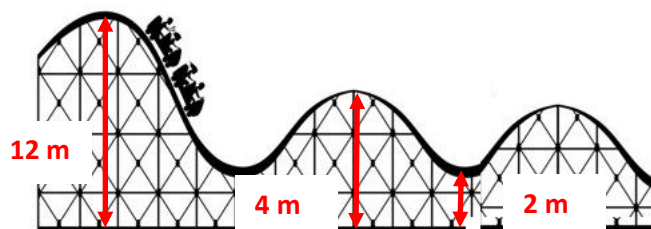
Find:

 - work done by person
 - work done by friction
 - final velocity of lawnmower
 - Find work done in stretching the spring whose force-distance graph is shown below
 - 0.0 cm to 4.0 cm
 - 0.0 cm to 10. cm
 - 4.0 cm to 10.0 cm
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8. a) A roller coaster car is at rest 12 m above ground, then moves through remainder of ride.

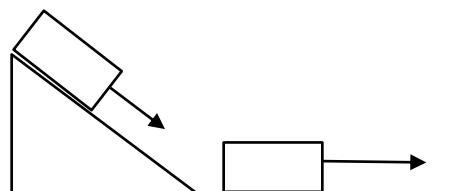
Assuming no energy loss, find speed of car at:

- i) 4 m above ground
- ii) 2 m above ground



- b) An object is thrown horizontally at 5.00 m/s off of a cliff that is 20.0 m high. Find object's speed as it impacts ground.

9. a) A 1.0 kg object is at rest 2.0 m above ground, then ends up at ground level moving 5.3 m/s as shown. Find energy lost in this process.



- b) A 22.0 kg child slides down a rough ramp that is 5.00 m high. At bottom of ramp, child has a speed of 2.50 m/s. Find energy lost as heat due to friction.

10. A person drags a 30.0 k g block across a horizontal floor ($\mu = 0.0200$) for 10 m over a period of 20.0 s. Find power expended in this process.

11. a) A 6.0 kg mass is pulled up a 10.0 metre long 8.75° frictionless incline at a constant speed of 1.50 m/s. Find: i) force required ii) work done iii) power required

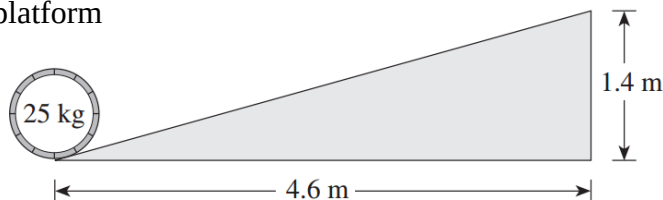
- b) A 1200 kg car can move up a frictionless 10.0° incline at a max speed of 15.0 m/s. At this speed, the drag force on the car is 7000 N. Find: i) force of propulsion from car's wheels ii) power of the car's engine

12. a) A climber of mass (80 kg) climbs from a height of 18 m to a height of 24 m. Climber expends 18 000 J of energy during this activity. Find efficiency of this process.

- b) A motor that is 82% efficient lifts a 1200 kg steel girder to the top of a 30.0 m high structure in 1.5 min. Find power rating of motor.

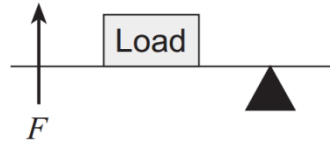
13. A barrel is rolled up a ramp to a loading platform with an effort force of 95 N as shown.

Find: a) ideal mechanical advantage
b) actual mechanical advantage
c) efficiency



14. For lever shown:

- effort (F) = 150.0 N, mass of load = 68.0 kg
- distance from load to fulcrum = 0.265 m
- distance from effort force to fulcrum = 2.10 m



Find: a) ideal mechanical advantage b) actual mechanical advantage c) efficiency

15. A mechanic used a pulley system to lift a 640 N engine using an effort force of 75 N.

- a) find actual mechanical advantage of scenario
- b) find how far down on rope mechanic would need to pull to lift engine up 21 cm if system was 85% efficient

16. A wheel-and-axle system is used to raise a 15 kg pail of water 4.5 m to the top of a well.

An average effort force of 11.2 N is used to turn the wheel (diameter = 38 cm) 70 times.

- a) Find efficiency of system using work (HINT: use circumference of wheel to calculate W_{in})
- b) Find axle diameter using mechanical advantage

KEY

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|-----------------|--|---------------------------------------|
| 1. a) 3.68 kJ | b) 459 N | |
| 2. a) 36 J | b) 3.8 m/s | |
| 3. a) 109 J | b) 85.3 J (F_F is opposite direction of motion) | c) 1.5 m/s |
| 4. a) 1.8 J | b) 4.2 J | c) 2.4 J |
| 5. a) 1.84 m | b) 9.07 m/s | c) 3.2 m |
| 6. 6.3 m/s | | |
| 7. a) 9.97 m/s | b) 18.8 m/s | c) 24.3 m/s |
| 8. a) i) 13 m/s | ii) 14 m/s | b) 20.4 m/s |
| 9. a) 5.6 J | b) 1010 J | |
| 10. 2.94 W | | |
| 11. a) i) 8.9 N | ii) 89 J | b) i) 9000 N ii) 136 kW = 181 hp |
| 12. a) 26% | b) 4.78 kW | |
| 13. a) 3.4 | b) 2.6 | c) 75% |
| 14. a) 7.92 | b) 4.44 | c) 56% |
| 15. a) 8.5 | b) 210 cm | |
| 16. a) 71% | b) 2.0 cm | |